CLOUDWATCH EVENTS, AND SNS

1. What are Events in AWS?

- "Amazon CloudWatch Events is a service that helps detect changes in AWS resources and respond to them automatically. — such as a file upload, a resource state change, or a scheduled time.
- It delivers real-time event streams which can trigger actions like invoking a Lambda function or sending a notification.
- It **supports both** AWS **service events** and custom **or scheduled events**, making it a powerful tool for building automated and event-driven architectures."

2. What is SNS in AWS?

- **Amazon SNS (Simple Notification Service)** is a **fully managed messaging service** used to **send notifications or messages** to multiple recipients at once.

[General Interview Questions]

Q1. What are CloudWatch Custom Metrics?

- **CloudWatch Custom Metrics** are **user-defined metrics** that you can create and publish to Amazon CloudWatch to monitor data that **is not available by default** from AWS services.

Q2. What are log groups and log streams in CloudWatch Logs?

- A **Log Group** is a **container for log streams** that share the same retention, monitoring, and access settings.
- A **Log Stream** is a **sequence of log events** from the same **source** (like an EC2 instance, Lambda function, or application).

Q3. How can you push logs from an EC2 instance to CloudWatch?

- Install and configure CloudWatch Agent, and assign an IAM role with permissions like logs:CreateLogGroup, logs:PutLogEvents.

Q4. What is the purpose of CloudWatch Events (EventBridge)?

- To **respond automatically** to changes in AWS resources or based on scheduled times **using rules and targets** like Lambda, Step Functions, or SNS.

Q5. How are CloudWatch Events different from Alarms?

- **Events:** React to AWS state changes or time
- **Alarms:** Trigger based on metric thresholds

Q6. Can you send the same SNS message to multiple email addresses?

- **Yes**, by subscribing multiple email endpoints to the same SNS topic.

Q7. How Do You Ensure Secure Delivery of Logs to CloudWatch?

1. Use **IAM Roles** and **Policies**
2. Encrypt Logs Using **KMS** (AWS Key Management Service)
3. Secure Communication with **HTTPS**
4. Use **VPC Endpoints**